AI-Assisted Software Development

Module 8a: Product Planning with AI Tools

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What is AI-Assisted Product Planning?

Traditional product planning can be:

- Time-consuming
- Prone to oversight
- Difficult to maintain consistency
- Challenging to keep organized

AI tools can help:

- Generate comprehensive documentation
- Maintain consistency
- Catch potential issues early
- Keep everything organized
- Provide different perspectives

Product Planning Workflow

Once requirements are gathered, the PRD is created:

…and then the technical planning begins:

Setting Up Your Planning Environment

1. Create a structured project directory:

```
project/
├── docs/
│   ├── prd.md
│   ├── requirements/
│   ├── architecture/
│   └── design/
├── src/
└── README.md
```

2. Use markdown files for:

- Documentation
- Requirements
- Technical specifications
- Design decisions

This structure helps AI tools maintain context throughout development.

Creating the PRD

Start with a clear template:

```
# Product Requirements Document

## Overview
[High-level product description]

## Problem Statement
[Clear problem definition]

## Target Users
[User personas and needs]

## Features
[Core functionality]

## Technical Requirements
[Implementation details]

## Constraints
[Limitations and boundaries]
```

Using AI for PRD Creation

Example conversation with AI:

Input to AI:

```
Create a PRD for a task management app with:
- Team collaboration
- Real-time updates
- Mobile support
- Integration with calendar
```

Key points:

- Start broad, then refine
- Ask for specific sections
- Use follow-up questions
- Keep context organized

PRD Best Practices

1. Be Specific
    - Clear feature descriptions
    - Measurable outcomes
    - Defined constraints
2. Maintain Context
    - Keep related files together
    - Use consistent terminology
    - Reference existing documentation
3. Iterate with AI
    - Start broad, then refine
    - Ask for different perspectives
    - Validate assumptions

Requirements Gathering

Turn client input into structured requirements:

1. Capture Raw Requirements

```
## Client Input
- "We need something fast"
- "Must work on phones"
- "Team features important"
```

2. Refine with AI

```
## Refined Requirements
- Response time < 500ms
- Progressive web app
- Real-time collaboration
```

3. Validate and Organize

```
## Categorized Requirements
Performance:
- Sub-second response time
- Offline capability

Platform:
- Cross-platform support
- Mobile-first design

Collaboration:
- Real-time updates
- Team permissions
```

Technical Specification Development

Use AI to convert requirements into technical specs:

Example:

```
Requirement: "Real-time collaboration"

Technical Specification:
- WebSocket implementation
- Event-driven architecture
- Conflict resolution system
- State synchronization
- Offline data handling
```

Documentation Organization

Keep documentation organized for maximum AI context:

1. Main Documents:
    - `prd.md`: Product Requirements
    - `architecture.md`: System Design
    - `api-spec.md`: API Documentation
2. Supporting Files:
    - `user-stories/`
    - `wireframes/`
    - `technical-specs/`

This organization helps the AI Tool maintain context for better assistance.

Using AI for Requirement Validation

Ask AI to analyze requirements for:

1. Completeness
    - Missing features
    - Edge cases
    - Dependencies
2. Consistency
    - Conflicting requirements
    - Technical feasibility
    - Resource constraints
3. Clarity
    - Ambiguous specifications
    - Undefined terms
    - Implicit assumptions

Context Management

Keep AI tools effective by:

1. Organizing Related Files

```
## Feature: User Authentication
- requirements/auth.md
- technical-specs/auth.md
- architecture/auth-flow.md
```

2. Using Clear References

```
See: [Authentication Flow](./auth-flow.md)
Related: [Security Requirements](./security.md)
```

3. Maintaining Documentation
    - Regular updates
    - Version control
    - Clear changelog

Recap

- Start with organized project structure
- Create comprehensive PRD with AI assistance
- Convert requirements to technical specifications
- Maintain clear documentation
- Use consistent organization
- Leverage AI for validation
- Iterate and refine continuously

Next Steps

- Review and refine PRD
- Begin technical planning
- Set up development environment
- Start UI/UX design process